Perfect play in a binary-scoring nonlocal game requires infinite dimension

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Abstract

We construct a finite rational two-player nonlocal game with a binary winning predicate that has a perfect strategy on $\ell^2 \otimes \ell^2$ and no perfect finite-dimensional strategy. The game has ten questions for Alice, eleven for Bob, at most six answers per question, and a uniform distribution on fifty-two question pairs. Two copies of the Flagged Magic-Square game are glued along complementary labels, with a boundary answer in the second copy. We prove the sharp dimension profile $D_\Gamma(\varepsilon) = 10\log_2(1/\varepsilon) + O(\log\log(1/\varepsilon))$ for arbitrary finite-dimensional local POVMs and mixed states. The lower bound is $10\log_2(1/\varepsilon) - 10\log_2\log_2(1/\varepsilon) - O(1)$. A dimension-independent square-root magic-square extraction gives a robust flagged sector law; the gluing then forces twenty Schmidt coefficients at every appreciable level of an approximate halving chain. The argument tracks Schmidt rank through measurement dilation. An explicit normalized spatial strategy has twenty coefficients at each geometric level. Its compression to $20K$ dimensions has loss exactly $1/[624(4^K - 1)]$, giving $D_\Gamma(\varepsilon) \le 10\log_2(1/\varepsilon) + 20$ for $0 < \varepsilon < 1$. Thus the finite-dimensional and spatial values are both one, but only the spatial value is attained.

1 The separation

A finite nonlocal game is specified by finite question and answer sets, a probability distribution π on question pairs, and a predicate V taking values in $\{0, 1\}$. A pure strategy consists of a unit vector $\psi \in H_A \otimes H_B$ and local positive operator-valued measurements (POVMs). It is *perfect* if its winning probability is one. We call it finite-dimensional when both local spaces are finite-dimensional, and spatial when they may be separable infinite-dimensional Hilbert spaces. The associated correlation sets are C_q and C_{qs}, respectively. Answer sets may depend on the question; padding by zero effects puts them in the usual fixed-alphabet convention.

Let $D_G(\varepsilon)$ be the least integer d for which a strategy on $\mathbb{C}^d \otimes \mathbb{C}^d$ with arbitrary POVMs has loss at most ε. Smaller spaces can be padded. Mixed shared states are also allowed in this definition. Throughout, dimension refers to this POVM convention.

Theorem 1.1. *There is a rational binary-scoring game Γ with ten Alice questions, eleven Bob questions, at most six answers per question, and a uniform distribution on fifty-two supported pairs, such that:*

1. *Γ has a normalized perfect strategy on $\ell^2 \otimes \ell^2$;*
2. *no finite-dimensional strategy is perfect;*

The authors developed this work with AI models from more than one developer and take responsibility for every claim. How the work was done, and what the machine checks cover, is described at the end of the paper.

3. for each integer $K \geq 1$, an explicit strategy of local dimension $20K$ has loss

$$\ell_{20K}^{\mathrm{cmp}} = \frac{1}{624(4^K - 1)}. \tag{1}$$

Both the finite-dimensional supremum and spatial value are one, but only the latter is attained, and

$$D_\Gamma(\varepsilon) \leq 20 \left\lceil \log_4 \left(1 + \frac{1}{624\varepsilon}\right) \right\rceil \qquad (\varepsilon > 0). \tag{2}$$

For $0 < \varepsilon < 1$ one also has $D_\Gamma(\varepsilon) \leq 10 \log_2(1/\varepsilon) + 20$.

Theorem 1.2 (Sharp dimension profile)**.** *There are constants $C > 0$ and $0 < \varepsilon_0 < 1/2$ such that every finite-dimensional strategy for Γ with loss at most ε, $0 < \varepsilon \leq \varepsilon_0$, using arbitrary local POVMs and a mixed or pure shared state, has each local dimension at least*

$$10 \log_2(1/\varepsilon) - 10 \log_2 \log_2(1/\varepsilon) - C. \tag{3}$$

Consequently,

$$D_\Gamma(\varepsilon) = 10 \log_2(1/\varepsilon) + O\big(\log\log(1/\varepsilon)\big) \qquad (\varepsilon \downarrow 0). \tag{4}$$

The lower bound controls all approximate strategies, through a robust sector law and a count of Schmidt multiplicities. The key analytic input is a dimension-independent square-root extraction for the row/column magic-square game, proved in Section 6. Sections 7 and 8 establish the robust sector law, gluing estimates and rank count. Local measurement dilation preserves the Schmidt rank used in this count, even when it enlarges the ambient Hilbert spaces.

Context and credit. Coladangelo and Stark separated finite-dimensional from spatial correlations with finite question and answer sets [3], using offset decompositions and Schmidt multisets. Beigi obtained such a separation with four binary measurements per party [1]. These are separations of exact correlations. Here every spatial correlation winning Γ perfectly lies outside C_q, so a finite game with a 0/1 winning predicate detects the distinction at perfect attainment. This answers the possibility raised in the second-page remark of Coladangelo–Stark [2] that a binary or integer-valued game score might not witness the separation. Binary scoring refers to the predicate, not to the number of answers.

Slofstra's nonclosure game [9] has a perfect approximate correlation but no perfect spatial strategy. His Theorem 3.2 equates finite and spatial perfect existence for linear-system games. Kim–Paulsen–Schafhauser prove $C_q^s(n,m) = C_{qs}^s(n,m)$ for synchronous correlations in their Theorem 3.10 [6]. These restricted classes do not include the present construction.

Our building block is Mančinska's Flagged Magic-Square game [8], including its calibrated frustrated bridge. Her Theorem 11 is a finite-dimensional exact self-test. We prove the sector identity on arbitrary Schmidt supports without appealing to a spatial extension of that theorem. The exact magic-square rigidity used below is Proposition 2 and Appendix A.2 of that paper, also proved here in the operator form we need. Lalonde's earlier game [7] requires nonmaximal entanglement for perfect play, but has a perfect strategy in local dimension six. The additional step here is the offset gluing that rules out every finite dimension.

Quantitative dimension witnesses predate this construction; for example, Slofstra–Vidick [10] prove polynomial dimension lower bounds for finite games. Douglas [5, Theorem 6] states a matching logarithmic law for the optimal-value deficit of I_{3322}. We do not use that paper's lower-bound certificates. The present theorem gives the sharp leading coefficient for this explicit spatially perfect, non-finitely-perfect game of value one. No absolute historical-priority claim is made.

2 The game and its finite building block

2.1 The flagged magic square

Write $\mathcal{R} = \{s \in \{\pm 1\}^3 : s_1 s_2 s_3 = 1\}$ and $\mathcal{C}_j = \{t \in \{\pm 1\}^3 : t_1 t_2 t_3 = \eta_j\}$, where $(\eta_1, \eta_2, \eta_3) = (1, 1, -1)$. Alice's questions are R_1, R_2, R_3, T_0, T_1; Bob's are C_1, C_2, C_3, U_0, U_1. Put

$$c_0 = r_0 = (1, 1, 1), \qquad c_1 = r_1 = (1, -1, -1), \qquad \sigma_{xy} = (-1)^{xy}.$$

The answers to R_i are $\mathcal{R} \sqcup \{f\}$, and those to C_j are $\mathcal{C}_j \sqcup \{f\}$. The answers to T_x are $\{+, -\} \sqcup (\mathcal{C}_1 \setminus \{c_x\})$, and those to U_y are $\{+, -\} \sqcup (\mathcal{R} \setminus \{r_y\})$. A sign answer is distinct from a sign triple. All twenty-five pairs have equal positive probability. The rules are:

1. At (R_i, C_j), (f, f) wins, a single flag loses, and nonflag answers (s, t) win exactly when $s_j = t_i$.
2. At (T_x, C_1), a sign wins exactly against f or c_x. A triple wins exactly against the identical triple.
3. At (R_1, U_y), a sign for Bob wins exactly against f or r_y. A triple wins exactly against the identical triple.
4. At (T_x, U_y), two signs $a, b \in \{\pm 1\}$ win exactly when $ab = \sigma_{xy}$. If at least one answer is a triple, they win.
5. All other pairs accept every answer pair.

This is G_{FMS}, in the conventions of [8].

Here is its ideal strategy. Let X, Y, Z be the Pauli matrices and use the square

$$(M_{ij}) = \begin{pmatrix} X \otimes I & I \otimes X & X \otimes X \\ I \otimes Z & Z \otimes I & Z \otimes Z \\ X \otimes Z & Z \otimes X & Y \otimes Y \end{pmatrix}. \tag{5}$$

Each row and column commutes; row products are I and column products are $I, I, -I$. Denote the rank-one joint spectral projections by $\pi^{\mathrm{row}}_{i,s}$ and $\pi^{\mathrm{col}}_{j,t}$. On $\mathbb{C}f \oplus \mathbb{C}^4$ take

$$\psi_* = \tfrac{1}{3}|ff\rangle + \tfrac{\sqrt{2}}{3} \sum_{z=0}^{3} |zz\rangle. \tag{6}$$

The row and column measurements consist of their spectral projections and the flag projection. Choose the distinguished eigenvectors as

$$|c_0\rangle = (1, 0, 1, 0)^T/\sqrt{2}, \qquad |c_1\rangle = (0, 1, 0, 1)^T/\sqrt{2},$$
$$|r_0\rangle = (1, 1, 1, 1)^T/2, \qquad |r_1\rangle = (1, -1, 1, -1)^T/2.$$

They satisfy $\langle c_x | r_y \rangle = \sigma_{xy}/\sqrt{2}$. The sign projections for T_x are onto $(|f\rangle \pm |c_x\rangle)/\sqrt{2}$, with the other column-one projections as the remaining outcomes. Bob uses $(|f\rangle \pm |r_y\rangle)/\sqrt{2}$ and the other row-one projections. All these projections are real. The row/column and calibration rules follow from their supports. For two signs the amplitude is proportional to $1 + ab\sigma_{xy}$, so the forbidden products vanish. Thus (6) wins perfectly.

2.2 Two copies and one label

Form Γ with Alice questions (h, q) and Bob questions (h, r), $h \in \{1, 2\}$, for gadget questions q, r. Bob has an additional question Lab with answers A, B. Copy one retains the gadget answers. Add a boundary answer $\perp$ to every question of copy two.

There are twenty-five internal pairs per copy and two label pairs:

$$((1, R_1), \mathsf{Lab}), \qquad ((2, R_1), \mathsf{Lab}).$$

Each of the fifty-two pairs has probability $1/52$. Copy one uses the gadget predicate. In copy two, $(\perp, \perp)$ wins, exactly one boundary answer loses, and otherwise use the gadget predicate. At the links accept exactly when

$$\begin{aligned} [a \neq f] &= [b = A] && \text{in copy one,} \\ [a \in \{f, \perp\}] &= [b = A] && \text{in copy two.} \end{aligned} \tag{7}$$

The first line concerns R_1, so $a \neq f$ means a magic-square answer. Unsupported pairs have probability zero. These rules specify a rational game with ten and eleven questions, five answers in copy one, six in copy two, and two label answers. The ancillary `game.json` records all supported pairs and forbidden answer pairs.

3 An inverse-free sector identity

Fix bases and identify ψ with its Hilbert–Schmidt coefficient operator $K : H_B \to H_A$. Bob's transposed effects are denoted by F. Then

$$p(E, F) = \|E^{1/2} K F^{1/2}\|_2^2, \qquad p(E, F) = 0 \implies EKF = 0. \tag{8}$$

Compress to the supports of the reduced densities. Now K is injective with dense range, and KK^*, K^*K are positive trace-class operators. Compression preserves correlations. No inverse of K will be used.

Lemma 3.1 (Checked classes)**.** *At a supported pair suppose the answers split into matching classes, with different classes always losing. In a perfect full-support strategy their coarse effects E_c, F_c are orthogonal projections, $E_c K = K F_c$, and E_c commutes with KK^*.*

Proof. The forbidden products and completeness give $E_c K = E_c K F_c = K F_c$. For $c \neq d$, $E_c E_d K = E_c K F_d = 0$. Dense range gives $E_c E_d = 0$, and completeness gives $E_c^2 = E_c$. Apply the same argument to K^* on Bob's side. Finally $E_c K K^* = K F_c K^*$ is self-adjoint, proving commutation. □

For orthogonal complete sectors satisfying $P_s K = K Q_s$, the compact operator K is the direct sum of its sector maps. Hence

$$\mathrm{Sch}(K) = \bigsqcup_s \mathrm{Sch}(P_s K Q_s). \tag{9}$$

Here Sch is the multiset of nonzero singular values with multiplicities. All sector vectors are unnormalized. Formula (9) holds also for countably infinite square-summable multisets, by the compact spectral theorem.

Lemma 3.2 (Exact magic-square algebra)**.** *Let reflections O_{ij} commute within rows and columns, with row products I and column products $I, I, -I$. There is a unitary identification with $\mathbb{C}^4 \otimes J$ carrying them to $M_{ij} \otimes I_J$ in* (5)*, for an arbitrary multiplicity space J.*

Proof. This is the operator rigidity underlying [8, Proposition 2 and Appendix A.2]. Put $a = O_{11}$, $b = O_{12}$, $c = O_{21}$, $d = O_{22}$. The cross commutations are $[a, b] = [a, c] = [b, d] = [c, d] = 0$. Parity gives $O_{13} = ab$, $O_{23} = cd$, $O_{31} = ac$, $O_{32} = bd$, and $acbd = O_{33} = -abcd$. Multiplication on the left by a and on the right by d gives $cb = -bc$. Using $[ac, bd] = 0$ and the cross commutations then gives $ad = -da$.

An anticommuting reflection pair is unitarily X, Z on $\mathbb{C}^2$ tensored with a multiplicity space: the second reflection exchanges the first's two eigenspaces. The second pair commutes with the first, so acts on its multiplicity space and permits the same construction. Taking (a, d) as the first-qubit (X, Z) and (b, c) as the second-qubit (X, Z) recovers (5). This works with infinite multiplicity. □

Proposition 3.3 (Flagged sector law)**.** *In every perfect finite-dimensional or spatial strategy for G_{FMS}, the flag and magic-square sector maps satisfy*

$$\mathrm{Sch}(K_m) = 4 \cdot (\sqrt{2}\ \mathrm{Sch}(K_f)). \tag{10}$$

Here $m \cdot S$ means m copies of S. The two sectors vanish together.

Proof. Flag consistency gives a common flag projection on each side, independent of the row or column, and $K = K_f \oplus K_m$. In the other sector, Lemma 3.1 makes the checked cell classes projections. A joint POVM with projection marginals is their joint spectral measurement: each fine effect is supported in the ranges specified by its marginals, and completeness recovers their products. Thus the cell reflections satisfy the row relations. Intertwining through K_m transfers Bob's column relations to Alice by dense-range cancellation, and conversely by injectivity. Lemma 3.2 on each side yields

$$K_m = I_4 \otimes D_m.$$

An intertwiner of the two full M_4 actions has this form by comparison of its sixteen matrix blocks. On its multiplicity supports D_m is injective with dense range.

Calibration identifies the sign supports as

$$S_x = P_f \oplus (|c_x\rangle\langle c_x| \otimes I), \qquad S'_y = Q_f \oplus (|r_y\rangle\langle r_y| \otimes I).$$

Put $X_x = E_x^+ - E_x^-$ and $Y_y = F_y^+ - F_y^-$, self-adjoint contractions on their sign supports. Identify each support with flag space direct sum multiplicity space and write

$$W_{xy} = S_x K S'_y = K_f \oplus \frac{\sigma_{xy}}{\sqrt{2}} D_m, \qquad H = K_f \oplus \frac{D_m}{\sqrt{2}}. \tag{11}$$

Each W_{xy} is injective with dense range. The two forbidden sign events imply

$$(I + X_x)W_{xy}(I - \sigma_{xy}Y_y) = 0,$$
$$(I - X_x)W_{xy}(I + \sigma_{xy}Y_y) = 0.$$

Their sum and difference give both identities

$$X_x W_{xy} = \sigma_{xy} W_{xy} Y_y, \qquad X_x W_{xy} Y_y = \sigma_{xy} W_{xy}. \tag{12}$$

Consequently $X_x^2 W_{xy} = W_{xy}$ and $W_{xy} Y_y^2 = W_{xy}$. Dense range and injectivity give $X_x^2 = Y_y^2 = I$. This proves projectivity of the unchecked signs with bounded operators alone.

For $y = 0$, $W_{00} = W_{10} = H$, so $X_0 = X_1 =: X$. For $y = 1$, $W_{01} = H$ and $W_{11} = \Delta_A H = H\Delta_B$, where $\Delta = I \oplus (-I)$ on each side. The intertwining equations imply $X = -\Delta_A X \Delta_A$. Thus

$$X = \begin{pmatrix} 0 & \beta^* \\ \beta & 0 \end{pmatrix}, \qquad \beta^*\beta = I, \quad \beta\beta^* = I.$$

The map β is unitary between flag and multiplicity spaces. From $XH = HY_0$ and self-adjointness follows $XHH^* = HH^*X$. Its off-diagonal blocks give

$$\beta K_f K_f^* = \tfrac{1}{2} D_m D_m^* \beta.$$

Their positive eigenvalues have this ratio. Taking square roots and counting the four copies in K_m proves (10). Unitarity of β also makes the sectors vanish together. □

This directly proves the spatial law without extending a finite-dimensional self-test by assumption. It makes no assertion about general commuting-operator strategies.

4 Gluing and the finite-dimensional obstruction

The support graph has Alice and Bob questions as its two vertex sets and supported pairs as edges; delete unused questions. Its connectedness is the hypothesis needed to propagate a common boundary projection.

Theorem 4.1 (Connected-support gluing)**.** *Let G be a finite rational game with connected support. At a supported pair (z, z'), suppose its answers split into consistent classes U, L. Assume every finite-dimensional perfect full-support strategy satisfies*

$$\mathrm{Sch}(K_U) = m \cdot \big(\kappa \,\mathrm{Sch}(K_L)\big), \qquad m \in \mathbb{N}_{\geq 1}, \quad \kappa > 1, \tag{13}$$

with both sectors nonzero. Suppose G has a perfect strategy with one Schmidt coefficient l in L and m coefficients κl in U.

Take two copies of the supported predicate, add the boundary answer to copy two, and link $(1, z), (2, z)$ to one Bob label question by $[a \in U] = [b = A]$ and $[a \in L \text{ or } a = \perp] = [b = A]$. Give all these pairs positive rational probabilities. The resulting game is spatially perfect, never finitely perfect, and has finite-dimensional supremum one. If (13) *holds spatially, the multiset identities below hold for every perfect spatial strategy.*

Proof. In copy two, boundary/nonboundary is a checked decomposition on every edge. Lemma 3.1 and connectedness propagate common projections $P_\perp, Q_\perp$ with $P_\perp K = K Q_\perp$. All other copy-two effects lie in their complements; conditioning on this complement when nonzero gives a perfect strategy for G.

Let $S_1 = \mathrm{Sch}(K_L^1)$ and $S_2 = \mathrm{Sch}(K_L^2)$ be the unnormalized lower sectors, and $T = \mathrm{Sch}(P_\perp K)$. The links give equalities of projections,

$$P_U^1 = P_L^2 + P_\perp, \qquad P_L^1 = P_U^2. \tag{14}$$

For example both sides of the first equality intertwine K with Bob's label-A projection, so dense range gives equality. Equations (9) and (13) imply

$$S_1 = m \cdot (\kappa S_2), \qquad m^2 \cdot (\kappa^2 S_2) = S_2 \sqcup T. \tag{15}$$

The nonboundary sector cannot vanish: the second link would kill copy one's lower sector, and its sector law would kill the whole state. Thus S_2 is nonempty. In finite dimension its least element $\mu > 0$ belongs to the left side of the second identity, so $\mu = \kappa^2 s$ for some $s \in S_2$. This contradicts $s = \mu/\kappa^2 < \mu$.

For spatial existence use basis vectors $A(i,j,k)$, $1 \le i,j \le m$, and $B(j,k)$, $1 \le j \le m$, $k \ge 0$. Give them respective Schmidt coefficients a_k, b_k, where

$$a_k = \kappa b_k, \qquad b_{k+1} = b_k/\kappa^2, \qquad b_0^2 = \frac{1-\kappa^{-4}}{m^2\kappa^2 + m}.$$

The squared coefficients sum to one. Copy one uses ideal blocks with $B(j,k)$ as L and $A(i,j,k)$ as U. Copy two uses $A(1,1,k+1)$ as L and $B(j,k)$ as U, since $b_k = \kappa a_{k+1}$. All other A vectors are boundary in copy two. Direct sums of ideal measurements, with the boundary projection, are bounded complete POVMs. Bob's label distinguishes A and B. Each internal pair wins on ideal blocks or boundary/boundary, and the two linked Alice projections equal the A projection. The diagonal state therefore wins both links. Finite truncation and compressed measurements give loss tending to zero by Lemma 5.1 below. □

Connectedness is explicit: on disconnected support, consistency gives one boundary projection per component and does not alone give a global projection. The concrete gadget has full pair support, so this issue does not arise for Γ.

Apply Theorem 4.1 with $m = 4$, $\kappa = \sqrt{2}$, $z = R_1$, $z' = C_1$, $U = \mathcal{R}$, $L = \{f\}$. The sector law is Proposition 3.3 and the ideal strategy is (6). In particular

$$S_1 = 4 \cdot (\sqrt{2}S_2), \qquad 16 \cdot (2S_2) = S_2 \sqcup T. \tag{16}$$

This proves finite impossibility without assuming positive boundary weight. In fact the weight is forced: copy one's law gives $\|S_1\|_2^2 = 1/9$, the second link gives $\|S_2\|_2^2 = 1/72$, and the first gives

$$\|T\|_2^2 = 8/9 - 1/72 = 7/8. \tag{17}$$

This mass calculation does not replace the minimum-coefficient argument.

5 Perfect spatial play and finite compression

5.1 The normalized state

For the concrete game index i,j by $0,1,2,3$. On each side use the basis $A(i,j,k), B(j,k)$, $k \ge 0$, and put

$$\begin{aligned} \psi = &\sum_{k\ge 0} \sum_{i,j=0}^{3} a_k |A(i,j,k)A(i,j,k)\rangle \\ &+ \sum_{k\ge 0} \sum_{j=0}^{3} b_k |B(j,k)B(j,k)\rangle, \\ b_k = \frac{2^{-k}}{4\sqrt{3}}, &\qquad a_k = \sqrt{2}b_k. \end{aligned} \tag{18}$$

The squared mass of level k is $16a_k^2 + 4b_k^2 = (3/4)4^{-k}$, so $\|\psi\| = 1$. Each local space is isomorphic to ℓ^2; there are two Hilbert tensor factors.

Copy one applies the ideal PVMs on each block

$$\big(B(j,k), A(0,j,k), A(1,j,k), A(2,j,k), A(3,j,k)\big).$$

Copy two applies them on

$$\big(A(0,0,k+1), B(0,k), B(1,k), B(2,k), B(3,k)\big),$$

and outputs $\perp$ on every remaining A vector. The coefficient ratio in each block is $(1, \sqrt{2}, \sqrt{2}, \sqrt{2}, \sqrt{2})$ since $b_k = \sqrt{2}a_{k+1}$. Bob labels the A and B subspaces. Alice's copy-one nonflag and copy-two flag-plus-boundary projections both equal the A projection; the diagonal state satisfies the links.

This covers all supported pairs. In copy one the nine row/column pairs, four calibration pairs and four sign pairs win by the ideal strategy; the remaining eight are free. In copy two all twenty-five pairs win on ideal blocks or boundary/boundary, with no mixed amplitude. Both links win as just shown. Thus (18) is a normalized perfect spatial strategy.

5.2 Loss under compression

Lemma 5.1 (Tail-to-loss bound)**.** *A perfect spatial strategy with Schmidt tail mass t outside a finite retained set has a normalized truncation with compressed measurements of loss at most t.*

Proof. Its loss operator satisfies $0 \le L \le I$ and $L\psi = 0$. The normalized truncation ϕ has overlap $\sqrt{1-t}$ with ψ. For $0 < t < 1$ write $\phi = \sqrt{1-t}\psi + \sqrt{t}\eta$ with $\eta \perp \psi$ unit. Then $\langle \phi, L\phi\rangle = t\langle \eta, L\eta\rangle \le t$. Compression of the local effects preserves this expectation and completeness. The zero-tail case is immediate. □

Retaining levels $0, \dots, K-1$ keeps $20K$ coefficients and mass $1 - 4^{-K}$. Compressed effects are POVMs; the sign effects at the incomplete last block need not be projections. The lemma gives

$$D_\Gamma(\varepsilon) \le 20\lceil \log_4(1/\varepsilon)\rceil \le 10\log_2(1/\varepsilon) + 20 \quad (0 < \varepsilon < 1).$$

The constant 10 uses arbitrary POVMs. A PVM-only convention requires a dilation and its dimension cost.

The compressed loss can be calculated exactly. All copy-one blocks remain intact. The only incomplete copy-two block is the last B quadruple, whose next-level flag is removed. Its unnormalized coefficient matrix is $\mathrm{diag}(0, b_{K-1}, b_{K-1}, b_{K-1}, b_{K-1})$. Only the four sign pairs lose. For forbidden answer signs u, v, $uv = -\sigma_{xy}$, and the probability is

$$\left| \frac{uv\, b_{K-1}}{2} \langle c_x | r_y \rangle \right|^2 = \frac{b_{K-1}^2}{8}.$$

Each of four pairs has two forbidden sign choices, giving total unweighted mass b_{K-1}^2. All row/column, calibration, boundary and link events still have zero loss. Dividing by 52 and the retained mass gives

$$\ell_{20K}^{\mathrm{cmp}} = \frac{b_{K-1}^2}{52(1-4^{-K})} = \frac{4^{-(K-1)}}{48 \cdot 52(1-4^{-K})} = \frac{1}{624(4^K - 1)}.$$

Solving for K gives (2) and completes Theorem 1.1. This is the loss of the specified compression; no optimality among all strategies of dimension $20K$ is asserted.

6 Quantitative consistency and magic-square extraction

In this and the next section Bob's operators are the physical operators on H_B, without transposition. Operators on opposite parties commute. For a vector v, $\mathrm{Sch}(v)$ now denotes its nonincreasing

Schmidt sequence, padded with zeros. Multiset unions are sorted before comparison. All vectors other than the original strategy vector may be unnormalized. Singular-value perturbation gives

$$\|\operatorname{Sch}(v) - \operatorname{Sch}(w)\|_2 \le \|v - w\|. \tag{19}$$

Indeed, the matrix singular-value inequality follows by applying the Hermitian eigenvalue inequality to the self-adjoint block matrices with off-diagonal blocks K, K^*; finite-rank approximation gives the compact Hilbert–Schmidt case. Sorting nonnegative sequences also cannot increase their Euclidean distance.

Lemma 6.1 (Approximate checked classes)**.** *For projective measurements, suppose mismatched answer classes at a pair always lose, with absolute losing mass at most η on a vector v. Their class projections satisfy $\|(E_c - F_c)v\| \le \sqrt{\eta}$. For the sign tests of G_{FMS}, writing $S_x = E_+^{T_x} + E_-^{T_x}$, $X_x = E_+^{T_x} - E_-^{T_x}$ and similarly S'_y, Y_y for Bob, one has*

$$\|(X_x Y_y - \sigma_{xy} S_x S'_y)v\| \le 2\sqrt{\eta}.$$

Proof. The squared norm of $(E_c - F_c)v$ is the sum of the probabilities of the disjoint mismatched-class events involving c. For signs, expand in the orthogonal vectors $E_a^{T_x} F_b^{U_y} v$ with $a, b \in \{+, -\}$. The coefficient $ab - \sigma_{xy}$ is zero on winning events and has magnitude two on losing events. □

Left multiplication by a contraction preserves a vector estimate. Right multiplication needs additional control. The following rule will provide it. If s are Alice reflections with Bob mirrors $\bar{s}$ satisfying $\|(s - \bar{s})v\| \le e$, then a word w of length l mirrors to the reversed Bob word with error at most le. Consequently, for an Alice operator R with $\|R\| \le 2$ and $\|Rv\| \le re$,

$$\|Rwv\| \le (r + 2l)e. \tag{20}$$

To see this, replace the rightmost letter by its mirror, commute that mirror past the remaining Alice letters, and continue. The resulting Bob word commutes with R and is unitary. The rule holds with the parties interchanged; the mirroring assertion also holds for contractions.

Theorem 6.2 (Square-root word extraction)**.** *Let $\|\phi\| \le 1$ and let local PVMs have valid magic-square row and column answers, with absolute losing mass at most η on each of the nine pairs. There are local isometries $V = V_A \otimes V_B$ and an auxiliary vector j such that, for every word w of length at most six in Bob's cell reflections and cell projections, including the empty word,*

$$\|Vw\phi - (I \otimes \widehat{w})(\Phi_4 \otimes j)\| \le C_{\mathrm{MS}}\sqrt{\eta}, \qquad C_{\mathrm{MS}} = 2^{24}. \tag{21}$$

Here $\Phi_4 = \frac{1}{2}\sum_{z=0}^{3} |zz\rangle$, and $\widehat{w}$ uses the ideal cells in (5)*, in the same order. The vector j need not be normalized. The constant is independent of dimension, also on separable Hilbert spaces.*

This statement is proved below for the actual row/column game. For comparison, the analytic estimate of Wu et al. [11, Appendix C, (124)] has a fourth-root rate in a correlation-error parameter; per-pair loss η gives checked correlation errors at most 2η. Coladangelo–Stark's magic-square statement is [4, Theorem 6.9], applying their general Theorem 4.16 to an equation/variable game. Their Definition 4.14 uses trace norm for the state and squared state-dependent operator distance. We do not use its asserted linear trace-norm rate: Lemma A.3 there fails already for $v_t = \sqrt{1-t}|0\rangle + \sqrt{t}|1\rangle$ and $P = |0\rangle\langle 0|$, since the squared projector defect is t but $\||v_t\rangle\langle v_t| - P\|_1 = 2\sqrt{t} > 6t$ for $0 < t < 1/9$.

Proof of Theorem 6.2. Let A_{ij}, B_{ij} be the cell reflections and put $e = 2\sqrt{\eta}$. Consistency gives $\|(A_{ij} - B_{ij})\phi\| \le e$. Alice's rows commute and have product I; Bob's columns commute and have products $I, I, -I$. Write $T \simeq_r U$ for $\|(T-U)\phi\| \le re$. Put

$$a = A_{11}, \quad b = A_{12}, \quad c = A_{21}, \quad d = A_{22}, \quad g = A_{31}, \quad h = A_{32}, \quad i = A_{33}.$$

Mirroring two letters to a Bob column, using its parity and mirroring back, gives

$$[a,c] \simeq_4 0, \qquad [b,d] \simeq_4 0,$$
$$g \simeq_3 ac, \qquad h \simeq_3 bd, \qquad i \simeq_3 -abcd.$$

Also $[a,b] = [c,d] = [g,h] = 0$ and $gh = i$ exactly. Substitution of h on the right, then of g through its two-letter suffix using (20), gives $gh \simeq_{10} acbd$. Thus $acbd \simeq_{13} -abcd$. Remove the left a and the right d, the latter by (20), to obtain $\{b,c\} \simeq_{15} 0$.

Similarly $hg \simeq_{10} bdac$, so $acbd \simeq_{20} bdac$. Commuting b, d through the suffix ac gives $bdac \simeq_8 dabc$; exact $[c,d] = 0$ followed by commuting b, d through c gives $abcd \simeq_6 adbc$. Combining these with the $13e$ estimate yields $(ad + da)bc \simeq_{47} 0$. Removing bc costs at most $4e$, so $\{a,d\} \simeq_{51} 0$. Transferring two-letter relations to Bob costs at most $4e$ more. Therefore $55e$ bounds all the corresponding commutation and anticommutation defects on both sides.

Consider the order-32 real Pauli group

$$G = \langle J, a, b, c, d : J^2 = a^2 = b^2 = c^2 = d^2 = 1,\ J \text{ central},$$
$$ad = Jda,\ bc = Jcb,\ \text{all other pairs commute} \rangle.$$

Every element has a unique form $J^z a^u b^v c^w d^x$ with binary exponents. Let f_A, f_B evaluate these forms in the physical generators with $J = -I$. Sorting a product involves at most eight letters and at most 32 adjacent exchanges. Each exchange costs at most $(55 + 16)e$ by (20), and cancellations are exact. In particular, on either party,

$$\|(f(g)f(h) - f(gh))\phi\| \le 4096e \qquad (g, h \in G). \tag{22}$$

Define isometries

$$W_A v = 32^{-1/2} \sum_{g \in G} |g\rangle \otimes f_A(g)v, \qquad W_B v = 32^{-1/2} \sum_{g \in G} |g\rangle \otimes f_B(g)v.$$

On the register use the right regular representation $(R_h F)(g) = F(gh)$. Squaring and averaging (22) over g gives

$$\|W_A f_A(h)\phi - R_h W_A \phi\| \le H, \qquad H = 4096e, \tag{23}$$

and likewise for Bob. An opposite-party isometry preserves these errors.

Since $f(Jg) = -f(g)$ exactly, each isometry has range in the register's $J = -1$ subspace. Every representation on this subspace is a multiple of the real four-dimensional representation $a = XI, b = IX, c = IZ, d = ZI$. For completeness, diagonalize the commuting c, d. The involution a exchanges the two d eigenspaces and preserves c; b exchanges the two c eigenspaces and preserves d. The four joint eigenspaces have equal multiplicity, and their identifications by a, b give precisely these Pauli actions. This also works for infinite multiplicity. Compose W_A, W_B with these identifications and call the result V.

Put $v = V\phi$. For each of the four generators, consistency and (23) give $\|(M_s^A - M_s^B)v\| \le 2H + e$. The four involutions $M_s^A M_s^B$ commute, with common $+1$ projection $P_0 = |\Phi_4\rangle\langle\Phi_4| \otimes I$. Hence

$$\|(I - P_0)v\|^2 \le \frac{1}{4} \sum_s \|(I - M_s^A M_s^B)v\|^2 \le (2H + e)^2.$$

Taking $\Phi_4 \otimes j = P_0 v$ proves the empty-word bound $\|v - \Phi_4 \otimes j\| \le 8193e$.

Every other Alice cell equals, or is within $3e$ of, its normal-form word of degree at most four. Its extraction error is thus at most $H + 3e + 8193e$. Bob's cells follow from consistency and $(M \otimes I)\Phi_4 = (I \otimes M^T)\Phi_4$, at a further cost of at most e. The common bound $2^{14}e$ covers a single reflection or projection on either side; a cell projection is $(I \pm M)/2$.

For a Bob word, mirror its last letter to Alice at cost at most e, commute it past the Bob prefix, apply the prefix estimate, and use the Alice single-letter estimate on $\Phi_4 \otimes j$. Passing from the physical isometric action to that target costs at most another $8193e$. Thus each added letter costs at most $e + 8193e + 2^{14}e$. For six letters, including the empty-word error, this is less than $2^{18}e$, which is covered by $2^{24}\sqrt{\eta}$. All estimates used absolute losing masses and $\|\phi\| \le 1$ throughout; there is no normalization factor or dimension dependence. □

7 The robust flagged sector law

We write $v \approx w$ in this section when $\|v - w\| \le C\sqrt{\eta}$, for a fixed constant independent of the strategy and its dimension. The constant may increase at successive steps: only finitely many contractions, bounded sums and words of bounded length occur.

Lemma 7.1 (Two-way spectral domination)**.** *Suppose local contractions satisfy $\|v - (A \otimes B)w\| \le d_1$ and $\|w - (A' \otimes B')v\| \le d_2$. Then*

$$\|\operatorname{Sch}(v) - \operatorname{Sch}(w)\|_2 \le \sqrt{d_1^2 + d_2^2}.$$

Proof. Let $u = \operatorname{Sch}(v)$, $t = \operatorname{Sch}(w)$, and let s, s' be the Schmidt sequences after the two contractions, respectively. Singular values decrease entrywise under contractions, so $s \le t$, $s' \le u$. Equation (19) gives $\|(u - t)_+\|_2 \le d_1$ and $\|(t - u)_+\|_2 \le d_2$. These positive parts have disjoint supports. The same argument for compact Hilbert–Schmidt maps follows by finite-rank approximation. □

Theorem 7.2 (Robust flagged sector law)**.** *There is an absolute constant C_{F} such that every projective G_{FMS} strategy on a vector ψ of norm at most one, with absolute losing mass at most η on each of its twenty-five pairs, satisfies*

$$\big\|\operatorname{Sch}(m) - 4 \cdot (\sqrt{2}\operatorname{Sch}(f))\big\|_2 \le C_{\mathrm{F}}\sqrt{\eta}, \quad f = E_f^{R_1}\psi, \quad m = \psi - f. \tag{24}$$

This holds in finite dimension and on separable spatial Hilbert spaces.

Proof. Put $P = E_f^{R_1}$, $Q = F_f^{C_1}$, and write P^i, Q^j for the other row and column flags. Let $H_y = E_{r_y}^{R_1}$, $G_x = F_{c_x}^{C_1}$. Cell projections, denoted $E_{i;j,s}, F_{j;i,s}$, sum the nonflag answers with cell sign s. Lemma 6.1 gives

$$\begin{aligned} P^i\psi \approx Q^j\psi \approx P\psi, \qquad & E_{i;j,s}\psi \approx F_{j;i,s}\psi, \\ Qm \approx P^i m \approx Q^j m \approx 0, \qquad & Pm = 0. \end{aligned} \tag{25}$$

The calibrations and signs give

$$S_x\psi \approx (Q + G_x)\psi, \quad S'_y\psi \approx (P + H_y)\psi, \quad X_Y Y_y\psi \approx \sigma_{xy}S_x S'_y\psi. \tag{26}$$

Mirrors and the nonflag game. If $L = F_{j;i,s}$ and $M = E_{i;j,s}$, then $L\psi \approx M\psi$. Moreover

$$PM\psi \approx PL\psi = LP\psi \approx LQ^j\psi = 0, \qquad MP\psi \approx MP^i\psi = 0.$$

Thus $Lm \approx Mm$. Replacing rightmost letters and commuting the opposite-party mirrors proves this assertion for every bounded-length Bob word, in reversed Alice order. The corresponding reflection estimate follows by bounded linear combinations.

Merge each row or column flag into a fixed valid nonflag answer, with the row-one target outside $\{r_0, r_1\}$ and the column-one target outside $\{c_0, c_1\}$. The merged measurements are PVMs with valid parities. For a nonflag losing event Λ, its amplitude on m differs from its amplitude on ψ by $\Lambda P\psi \approx \Lambda P^i\psi = 0$. Events involving merged flags have amplitudes $O(\sqrt{\eta})$ by (7). There are a fixed number of outcomes, so the losing *mass* on m is $O(\eta)$ on every pair. Theorem 6.2 therefore applies at square-root rate. An unmerged word differs from the merged word by a bounded sum of terms containing a column flag. Mirror the suffix to Alice, commute it past that flag and use $Q^j m \approx 0$. Each term is $O(\sqrt{\eta})$. Consequently the extraction applies to the unmerged words below as well.

The needed ideal words. Let

$$\widetilde{B}_y = F_{3;1,(r_y)_3}F_{2;1,(r_y)_2}F_{1;1,(r_y)_1}.$$

It mirrors H_y on m. Define the physical reflections

$$A = E_{1;2,+} - E_{1;2,-} + P, \quad B_{12} = F_{2;1,+} - F_{2;1,-} + Q^2, \quad B = F_{1;2,+} - F_{1;2,-} + Q.$$

In the ideal square $A = IX$, $B = IZ$. The exact ideal identities are

$$\begin{aligned} &\widetilde{B}_0G_1 = \widetilde{B}_0G_0B_{12}, && \widetilde{B}_1G_1 = -\widetilde{B}_1G_0B_{12}, \\ &\widetilde{B}_1G_0 = B\widetilde{B}_0G_0. \end{aligned} \tag{27}$$

Indeed, $IXc_0 = c_1$, $IXr_y = (-1)^y r_y$, $IZr_0 = r_1$ and $IZc_0 = c_0$. The reversed order under Bob's transposed action on Φ_4 is essential. A column answer projector uses two cell projections because of parity, so the longest word displayed has length six. Theorem 6.2 transports these identities to m. For its auxiliary vector j the same theorem and singular-value perturbation give

$$\mathrm{Sch}(m) \approx 4 \cdot (\mathrm{Sch}(j)/2), \qquad \mathrm{Sch}(\widetilde{B}_0G_0m) \approx \mathrm{Sch}(j)/(2\sqrt{2}). \tag{28}$$

For the second equality, the product of the rank-one row and column projections has singular value $|\langle c_0|r_0\rangle| = 1/\sqrt{2}$; the factor $1/2$ comes from Φ_4.

Frustration on the sign supports. Put $z_{xy} = S_xS'_ym$, $u = S_0S'_0\psi$ and $\Delta_A = 2P - I$, $\Delta_B = 2Q - I$. Calibration implies $S_xf \approx S'_yf \approx f$. For example, $S_xP\psi \approx S_xQ\psi = QS_x\psi \approx Q(Q + G_x)\psi \approx f$. Using mirrors on m then gives

$$z_{xy} \approx \widetilde{B}_yG_xm, \qquad Pz_{xy} \approx Qz_{xy} \approx 0, \qquad S_xS'_y\psi \approx f + z_{xy}. \tag{29}$$

For P annihilation commute P through the Bob word and use $Pm = 0$. For Q annihilation first mirror the *whole* Bob word to Alice, then commute Q through that word and use $Qm \approx 0$. The three identities (27), consistency $Am \approx B_{12}m$, and $Af = f$, $Bf \approx f$ yield

$$S_1S'_0\psi \approx Au, \quad S_0S'_1\psi \approx Bu, \quad S_1S'_1\psi \approx AB\Delta_Au. \tag{30}$$

Here A commutes with P exactly, and $\Delta_A u \approx \Delta_B u$ by both annihilations in (29).

Set $X' = AX_1A$, $Y' = BY_1B$. Substituting (30) in the four sign tests and conjugating by the indicated reflections gives

$$X_0Y_0u \approx u, \quad X'Y_0u \approx u, \quad X_0Y'u \approx u, \quad \Delta_A X'\Delta_A Y'u \approx -u. \tag{31}$$

Subtract the first two equations and apply Y_0, using $Y_0^2u = u$, to get $(X_0 - X')u \approx 0$. This also holds on $\Delta_A u$: replace $\Delta_A u$ by $\Delta_B u$ and commute Δ_B past the Alice operators. Replace X' in the fourth equation and add the third. To remove Y', note that

$$Y'^2u = BS_1'Bu \approx BS_1'(S_0S_1'\psi) = B(S_0S_1'\psi) \approx u.$$

It follows that

$$(X_0 + \Delta_A X_0 \Delta_A)u = 2(PX_0P + P^\perp X_0 P^\perp)u \approx 0.$$

Since $u \approx f + z_{00}$ and $Pz_{00} \approx 0$, the two orthogonal range blocks imply $PX_0f \approx 0$ and $P^\perp X_0 z_{00} \approx 0$. Projecting $X_0Y_0u \approx u$ onto these ranges, and using that Y_0 commutes with P, proves the two directions

$$X_0Y_0f \approx z_{00}, \qquad X_0Y_0z_{00} \approx f. \tag{32}$$

For example the P projection identifies $PX_0Y_0z_{00}$ with f, and its complementary component is small by the preceding block estimate; the other direction follows from the complementary projection.

Spectra. The operators X_0, Y_0 are contractions, so Lemma 7.1 gives $\mathrm{Sch}(f) \approx \mathrm{Sch}(z_{00})$. Combine (29) and (28): four copies of $\mathrm{Sch}(j)/2$ are four copies of $\sqrt{2}\,\mathrm{Sch}(j)/(2\sqrt{2})$. This proves (24). All operations have dimension-independent bounds and bounded word lengths. They establish a finite absolute C_{F}; no useful numerical value for this constant is asserted. □

8 Gluing estimates and logarithmic rank

Lemma 8.1 (Approximate gluing)**.** *There is a constant C_{G} such that a unit-vector PVM strategy for Γ of loss at most ε has the following properties. Let L_A, L_B be Bob's two label-answer projections, put*

$$\alpha = L_A\psi, \quad \beta = L_B\psi, \quad \phi_2 = E_f^{(2,R_1)}\psi, \qquad c = \mathrm{Sch}(\alpha), \quad b = \mathrm{Sch}(\beta), \quad a = \mathrm{Sch}(\phi_2).$$

The subscripts on L_A, L_B denote answers, not tensor factors. With $P_\perp = E_\perp^{(2,R_1)}$, $Q_\perp = F_\perp^{(2,C_1)}$, $t = \mathrm{Sch}(P_\perp Q_\perp \psi)$ and $\delta = C_{\mathrm{G}}\sqrt{\varepsilon}$,

$$\|c - (a \sqcup t)\|_2 \le \delta, \tag{i}$$

$$\|c - 4 \cdot (\sqrt{2}b)\|_2 \le \delta, \tag{ii}$$

$$\|b - 4 \cdot (\sqrt{2}a)\|_2 \le \delta, \tag{iii}$$

$$\|\,\mathrm{Sch}(\psi) - (c \sqcup b)\|_2 \le \delta, \tag{iv}$$

$$\|\alpha\|^2 + \|\beta\|^2 = 1. \tag{v}$$

Proof. Every supported pair has losing mass at most 52ε. Link one identifies the first-copy magic-square vector with α and its flag vector with β, with vector error $O(\sqrt{\varepsilon})$. Apply Theorem 7.2 and singular-value perturbation to obtain (ii).

For (iii) remove the boundary coherently: put $\psi' = (I - P_\perp)\psi$. Boundary consistency on all twenty-five copy-two pairs identifies each boundary vector with $P_\perp\psi$ within $O(\sqrt{\varepsilon})$. In particular every row or column boundary projection applied to ψ' is $O(\sqrt{\varepsilon})$. Merge the boundary into a valid retained answer, choosing a magic-square answer at a row or column and an ordinary triple at a sign question. For a nonboundary losing projector Λ, $\Lambda P_\perp\psi$ is small: replace $P_\perp$ by the boundary projection at the relevant opposite-party question, where its product with Λ is zero. Thus every losing amplitude on ψ' is $O(\sqrt{\varepsilon})$ and the merged FMS losing mass is $O(\varepsilon)$. At R_1 its flag vector is exactly ϕ_2, and its nonflag vector is exactly $E_{\mathrm{MS}}^{(2,R_1)}\psi$, which link two identifies with β. The robust sector law proves (iii).

For (i), link two identifies α with $(E_f^{(2,R_1)} + P_\perp)\psi$. At the pair $((2,R_1),(2,C_1))$ the three classes flag, boundary and magic square are consistent, so this vector is within $O(\sqrt{\varepsilon})$ of

$$w = E_f^{(2,R_1)} F_f^{(2,C_1)}\psi + P_\perp Q_\perp \psi.$$

Its two summands are orthogonal blocks on *both* parties, hence their Schmidt lists form the union $\mathrm{Sch}(w)$. The first block is within $O(\sqrt{\varepsilon})$ of ϕ_2. Singular-value perturbation and sorting prove (i). For (iv) use the two-sided block vector

$$E_{\mathrm{MS}}^{(1,R_1)} L_A\psi + E_f^{(1,R_1)} L_B\psi.$$

Consistency makes this vector close to ψ and its blocks close to α, β, respectively. Finally (v) is exact completeness of the label PVM. A single enlarged constant covers all four errors. □

Lemma 8.2 (Descent with multiplicities)**.** *Suppose finitely supported nonnegative sorted sequences a, b, c, t satisfy (i)–(iii) in the infinity norm with error at most $\delta > 0$. Let $\mu_0 = \max a$. For every integer $k \geq 0$ with $\mu_0 2^{-k} \geq 64\delta$ there are distinct positive entries $\mu_k = a_{i_k}$ such that*

$$|\mu_k - \mu_0 2^{-k}| \leq (2 + \sqrt{2})\delta. \tag{33}$$

If K is the number of these integers, then $K \geq \log_2(\mu_0/(64\delta))$. At each index i_k, four entries of b are within δ of $\sqrt{2}\mu_k$ and sixteen entries of c are within $(1 + \sqrt{2})\delta$ of $2\mu_k$. These are disjoint index blocks for distinct k.

Proof. For a positive entry μ of a, select its occurrence p in the sorted union $a \sqcup t$, retaining occurrences when values coincide. Read (i) and (ii) at p to obtain $|b_{\lceil p/4 \rceil} - \mu/\sqrt{2}| \leq \sqrt{2}\delta$. Read (iii) at $\lceil p/4 \rceil$ to find an entry of the zero-padded a within $g_0 = (1 + 1/\sqrt{2})\delta$ of $\mu/2$. Iteration contracts the preceding error by $1/2$, so its total is at most $g_1 = 2g_0 = (2 + \sqrt{2})\delta$. Retain exactly the integers in the statement. Their entries are positive, so none is a padded zero. Consecutive entries differ by at least $\mu_0 2^{-k}/2 - 2g_1 > 0$. Thus their a indices are distinct. If $\mu_0 \geq 64\delta$, the number retained is $\lfloor \log_2(\mu_0/(64\delta)) \rfloor + 1$; if not, take $K = 0$. Both conventions give the stated inequality.

Relation (iii) supplies the four b indices $4(i_k - 1) + 1, \ldots, 4i_k$. Relation (ii) supplies the sixteen c indices $16(i_k - 1) + 1, \ldots, 16i_k$, with the stated accumulated error. Distinct i_k give disjoint blocks, regardless of equal spectral values. □

Proof of Theorem 1.2. First consider a finite-dimensional mixed-state POVM strategy on local spaces of dimensions d_A, d_B. Loss is linear in the state, so some pure component has loss at most ε using the same measurements. Let its Schmidt rank be $r \leq \min(d_A, d_B)$. Give the finitely many local POVMs common Naimark dilations with fixed pure local ancillas. The resulting PVM strategy preserves the correlations and the Schmidt rank r. The dilated ambient dimensions may be larger

and are never used as the original dimensions. Every one-sided projection and local contraction used below has Schmidt rank at most r.

Apply Lemma 8.1. Write $x = \|a\|_2$, $y = \|b\|_2$, $z = \|c\|_2$. Its relations give

$$z \le \sqrt{8}y + \delta, \quad y \le \sqrt{8}x + \delta, \quad y^2 + z^2 = 1.$$

Since $y^2 + (\sqrt{8}y + \delta)^2 \le (3y + \delta)^2$,

$$y \ge (1 - \delta)/3, \qquad x \ge (1 - 4\delta)/\sqrt{72}. \tag{34}$$

Choose $\varepsilon_0 > 0$ so that $C_{\mathrm{G}}\sqrt{\varepsilon_0} \le 1/100$. Then $x > 0.11$. As $\operatorname{rank}(\phi_2) \le r$, $\mu_0 = \max a \ge 0.11/\sqrt{r}$.

The 2-norm estimates imply infinity-norm estimates, so Lemma 8.2 applies. All its selected b and c entries exceed δ: even the smallest chain entry is at least $(64 - 2 - \sqrt{2})\delta$. Relation (iv) then forces at least $20K$ positive entries in $\mathrm{Sch}(\psi)$. Indeed, match occurrences in the sorted union $c \sqcup b$; an occurrence larger than δ cannot match a zero with coordinate error at most δ. Multiplicity cannot be lost by coincident values. Consequently

$$\begin{aligned} r \ge 20K &\ge 20 \log_2 \frac{\mu_0}{64 C_{\mathrm{G}} \sqrt{\varepsilon}} \\ &\ge 10 \log_2(1/\varepsilon) - 10 \log_2 r - 20 \log_2(64 C_{\mathrm{G}}/0.11). \end{aligned} \tag{35}$$

Put $L = \log_2(1/\varepsilon)$. If $r \ge 10L$ the desired lower bound is immediate. Otherwise $\log_2 r \le \log_2 L + \log_2 10$ in (35). Absorb the fixed constants to get $r \ge 10L - 10 \log_2 L - C$. This bounds each original local dimension. Together with Theorem 1.1 it proves the assertion for D_Γ and all the stated quantifiers. At zero loss the same descent has arbitrarily many positive entries; one treats that case directly, without taking a logarithm of zero. □

9 Exact tails and remaining questions

Proposition 9.1 (Exact perfect-state tails)**.** *Every normalized perfect spatial strategy has a nonempty sector multiset S_2 closed under halving. If $s_0 = \max S_2$, its Schmidt tail after deleting the r largest coefficients satisfies*

$$t_\psi(r) \ge 96 s_0^2 \, 4^{-\lceil r/20 \rceil} \ge 24 s_0^2 \, 2^{-r/10}. \tag{36}$$

Proof. The spatial sector law gives (16) for compact sector maps. If $s \in S_2$, the second identity puts s in $16 \cdot (2S_2)$, hence $s/2 \in S_2$. Square summability gives a largest coefficient $s_0 > 0$ and thus the distinct chain $s_0 2^{-k}$.

The full Schmidt multiset contains sixteen copies of $2s_0 2^{-k}$ from copy one's upper sector and four copies of $\sqrt{2} s_0 2^{-k}$ from its lower sector, for every k. This selected submultiset has squared mass $72 s_0^2 4^{-k}$ per level, and tail $96 s_0^2 4^{-K}$ after K full levels. Deleting r coefficients from the full multiset cannot leave less squared mass than optimally deleting r from this submultiset. Deleting $20\lceil r/20 \rceil$ selected coefficients can only decrease the latter tail. This proves the first inequality; $\lceil r/20 \rceil \le r/20 + 1$ gives the second. □

No nonzero finite-rank projection can commute with every Alice effect in a perfect full-support strategy. Projecting the vector onto it preserves all losing zeros and gives a nonzero finite-Schmidt-rank perfect vector. Bob can then be compressed to its finite Schmidt support, contradicting Theorem 1.1.

Proposition 9.1 concerns exact perfect states. It does not compare arbitrary approximate strategies with truncations of perfect states or prove extremality of (18) for truncation. Such a comparison requires a separate argument.

Corollary 9.2 (Profile at dyadic accuracy)**.** *For* $A_\Gamma(n) = \log_2 D_\Gamma(2^{-n})$ *one has*

$$A_\Gamma(n) = \log_2 n + \log_2 10 + O\big((\log n)/n\big) \qquad (n \to \infty).$$

Proof. Theorems 1.1 and 1.2 give $D_\Gamma(2^{-n}) = 10n + O(\log n)$. Take logarithms. □

Remaining questions. The logarithmic dimension law is established. It remains open whether the $O(\log\log(1/\varepsilon))$ remainder in (4) can be replaced by $O(1)$, and whether the specified compressions are optimal at their finite dimensions. More generally, which finite spatially perfect games admit matching quantitative sector laws? The exact tail result alone does not answer these approximation questions.

How this work was done

The authors developed the construction and proofs with AI models from more than one developer. The proofs were checked against the cited primary statements and by finite symbolic and numerical calculations. The square-root word extraction is proved here directly, with its norm convention, constant and row/column hypotheses stated explicitly. Nonexistence in finite dimension follows from the sector-and-minimum proof; numerical optimization is not a certificate of nonexistence.

The ancillary package contains the rational predicate, a deterministic generator, symbolic checks, finite compression checks, recorded outputs, a sequential runner, and SHA-256 checksums. The symbolic checks verify all forbidden gadget products exactly, projectivity, completeness, overlap signs, normalization and the incomplete-block loss. Numerical checks cover every supported pair for the specified compressions and compare their loss with (1). Positivity tolerances are supplemental; positivity follows analytically by compression of projections. Additional exact checks cover the three bridge words, the order-32 real Pauli group and the trace-norm counterexample. Numerical checks test regular-representation extraction, complex local perturbations and sector defects, with one solver thread. A separate descent check verifies thresholds and multiplicities. These finite tests do not establish a uniform lower bound; the analytic sector, gluing and rank proofs do. The infinite strategy is justified by the normalized direct-sum construction. No proof-assistant certification or exhaustive historical-priority search is claimed.

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